Laasya M. Koduru

669-246-2174 laasya.koduru@gmail.com lmkoduru.github.io Santa Barbara, CA 93106

Education

University of California, Santa Barbara — Santa Barbara, California
Doctor of Philosophy (Ph.D.), Computer Science — Sept 2024–Present
Advisors: Dr. Arpit Gupta, Dr. Elizabeth Belding

University of California, Santa Barbara — Santa Barbara, California
Master of Science (M.S.), Computer Science — Sept 2023–Present

University of California, Santa Cruz — Santa Cruz, California
Bachelor of Science (B.S.), Applied Mathematics — Sept 2020–June 2023

Research Interests

LLM and computer-use agents that automate computer workflows, where the apps and interfaces they operate keep changing. Interested in how deep networks forget, adapt, and lose plasticity under that kind of non-stationary data.

Experience

Idaho National Laboratory — Idaho Falls, ID
Cybersecurity Machine Learning Algorithms Intern — June–Sept 2025

- Developed a machine learning algorithm that separates mixed Ethernet signals into their original source signals, reaching 96% accuracy against a 90% baseline.
- Work adds new capabilities for securing control systems in critical infrastructure. Patent application under review.

University of California, Santa Barbara — Santa Barbara, CA
Graduate Student Researcher — Sept 2023–Present
Systems & Networking Lab (SNL) and Mobility Management & Networking Lab (MOMENT)

- Developed BQT+, an agentic framework to automate user interactions with websites to gather public broadband data. Scaled data collection across multiple ISPs with a containerized Docker system using PyAutoGUI.

Projects

CVE-Genie: Automated CVE Reproduction | *Python* — Sept–Dec 2024

- Co-built an LLM multi-agent system that rebuilds, exploits, and verifies CVEs from public reports.
- Reproduced 50 CVEs, including C/C++ memory-safety bugs.

MOVERC: MongoDB Version Classifier | *Python* — April–June 2024

- Built a random forest that infers MongoDB versions (3.0–7.0) from packet headers to flag vulnerable servers.
- Generated 30K flows from 220 MongoDB apps under varied network conditions (netMosaic, NTLFlowLyzer).
- Reached 0.79 precision, 0.74 weighted F1 (vs. 0.53, 0.42 for a neural net); audited via Trustee and ablation.

CodeInsight | *Java* — March 2024

- LLM tool built on GPT and the VS Code Java debugger that answers "why" questions about program output.

PokePortal | *Ruby, React, JavaScript, HTML, CSS* Nov 2023–Dec 2023

– Social gaming web app for scheduling battles with friends, following live Pokémon news, and posting comments.

Movie Recommendation System | *Python* April 2023–June 2023

– Developed with Pearson correlation and a user-user similarity matrix that generates movie recommendations for users leveraging their past rated movies, submission date, and movie genre.

Technical Skills

- **Security:** Network traffic analysis, vulnerability reproduction, exploit verification
- **Languages:** Python, C, C++, Java
- **Machine Learning:** LLM Agents, PyTorch, Keras, TensorFlow
- **Systems & Tools:** Docker, Git, AWS, SQL, PyAutoGUI, LATEX

Selected Publications

[1] Jaber Daneshamooz, Eugene Vuong, **Laasya Koduru**, Sanjay Chandrasekaran, Arpit Gupta, "NetGent: Agent-Based Automation of Network Application Workflows", Workshop on Machine Learning for Systems at NeurIPS, Dec 2025.

[2] **Laasya Koduru**, Sylee Beltiukov, Alexander Nguyen, Eugene Vuong, Jaber Daneshamooz, Tejas Narechania, Elizabeth Belding, Arpit Gupta, "Robust and Extensible Measurement of Broadband Plans with BQT+", arXiv preprint, Feb 2026.

[3] Haarika Manda, Varshika Srinivasavaradhan, **Laasya Koduru**, Kevin Zhang, Xuanhe Zhou, Udit Paul, Elizabeth Belding, Arpit Gupta, Tejas Narechania, "The Efficacy of the Connect America Fund in Addressing US Internet Access Inequities", Proceedings of the ACM SIGCOMM 2024 Conference. Full list at lmkoduru.github.io.

Honors

- UC Santa Barbara Neal Fenzi Resonant Founder Fellowship April 2026–Present
- The National GEM Consortium Ph.D. Science and Engineering Fellowship April 2025–Present
- IETF/IRTF Applied Networking Research Prize April 2025
- UC Santa Barbara Computer Science Summer Fellowship June 2024

Leadership & Service

UCSB CS Grad Student Association (GSA)

Vice President, Secretary Sept 2025–Present

– Represent CS graduate students at campus-wide GSA meetings; serve on the UCSB AI committee, speak at open houses, plan events.

Early Research Student Program, UC Santa Barbara

Ph.D. Mentor Sept 2025–Present

– Mentor a team of undergraduate students through an end-to-end research project, guiding research methodology and technical execution.